

Artificial Intelligence in the Contemporary World

Project Title:
Banking Customer Account Dashboard and Financial Health Analysis

Project Domain

Excel and Power BI with Copilot

A Collaborative Initiative

Equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

Banks and financial institutions generate large volumes of customer data every day including demographic information, financial activity, product usage, and account history. However, much of this data remains underutilized because it is often unstructured, inconsistent, or not analyzed effectively. Without proper analytics and visualization, organizations struggle to identify meaningful insights that can support strategic business decisions.

One of the major challenges faced by banks today is customer churn. Customer churn occurs when customers discontinue their relationship with the bank, leading to loss of revenue and increased customer acquisition costs. Identifying the factors that influence churn is therefore critical for maintaining profitability and customer loyalty.

The dataset used in this project contains 10,000 customer records with attributes such as credit score, geography, age, tenure, account balance, number of products, active member status, and estimated salary. The target variable “Exited” indicates whether a customer left the bank.

This project aims to analyze the dataset using Excel and Power BI to identify patterns that influence customer churn and financial behavior. Artificial Intelligence tools such as Microsoft Copilot are also used to assist in analysis, insight generation, and documentation. The insights derived from this analysis will help organizations design effective customer retention strategies and improve overall financial performance.

4. Objectives

- To collect and explore the customer churn dataset and understand its structure and attributes.
- To clean and preprocess the dataset by handling duplicates, validating fields, and ensuring data accuracy.
- To analyze customer behavior patterns using Excel formulas, pivot tables, and Power BI analytics.
- To identify the key factors influencing customer churn in the banking sector.
- To calculate important business KPIs such as churn rate, retention rate, and revenue at risk.
- To build an interactive dashboard that presents financial and customer insights visually.
- To integrate AI tools such as Microsoft Copilot to enhance analysis and improve productivity.
- To generate actionable insights that help banks reduce churn and improve customer retention.

5. Methodology / Implementation Plan

The project follows a structured data analytics workflow consisting of five major phases.

#	Phase	Description
1	Data Collection & Exploration	Obtain the dataset from Kaggle and understand field names, data types, and dataset structure.
2	Data Cleaning & Preprocessing	Remove duplicates, verify categorical values, and validate numerical fields.
3	Analysis & AI Integration	Apply Excel formulas, pivot tables, and Power BI analytics to generate insights.
4	Visualization / Output	Create interactive dashboards with charts, KPIs, and filters in Power BI.
5	Documentation & Submission	Compile project documentation, dashboard outputs, and analysis results for submission.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version
- Microsoft Copilot or AI Chatbot tools
- Internet connection and web browser

B. Dataset

The dataset used in this project contains 10,000 customer records representing bank customers. Each record includes attributes such as credit score, age, geography, gender, tenure, account balance, number of products, active member status, estimated salary, and the churn indicator 'Exited'.

The dataset helps analyze customer financial health, identify patterns of churn behavior, and understand how customer attributes influence retention.

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	Week 1
2	Data cleaning and preprocessing	Week 2
3	Analysis and AI integration	Week 2 – 3
4	Dashboard build and visualization	Week 3
5	Report writing and submission	Week 4

7. Expected Outcomes / Conclusion

The project will produce a clean and validated dataset with documented preprocessing steps. It will identify key drivers of customer churn and evaluate important KPIs including churn rate, retention rate, active member percentage, and revenue at risk.

An interactive Power BI dashboard will visually communicate insights and support data-driven decision making. The project also demonstrates how AI tools such as Microsoft Copilot can assist analysts in generating summaries, automating repetitive tasks, and improving analytical efficiency.

The insights generated from this analysis can help banks design targeted retention strategies, improve customer satisfaction, and minimize revenue loss.

8. References

- Edunet Foundation – AICW Fellowship Program Guidelines (2026)
- Microsoft Excel Documentation – Microsoft Support
- Microsoft Power BI Documentation – Microsoft Learn
- Kaggle – Customer Churn Dataset Repository